SMOOTHNESS OFF V(I) VERSUS RIG-SMOOTHNESS OF THE COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 955-1013, following the statement at 942-953.

Editorial note. Both 'small detail omitted' flags are readings through lemma-equivalent-with-artin-smooth, whose use 'without further mention' the author declares at the start; flagged in the notes.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 967, 1007 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Equivalent conditions for rig-smoothness. lemma-equivalent-with-artin-smooth (same chapter, used without further mention; its proof has stated omissions and was not counted). Translates between annihilated Ext classes and the dotted-arrow condition.
- Completion of the naive cotangent complex. lemma-NL-is-completion (same chapter, cited at ~965) and More on Algebra lemmas base-change-property-ext-1-annihilated, base-change-RHom (cited at ~966-982). Relates NL of B and of the completion.
- Nakayama and Krull tools. Algebra, Lemma algebra-lemma-NAK (cited at ~991). Produces the element g = 1 + x annihilating the class module.

Source reference. Original reference 00DV: focused-seeds/geometry/S9-restricted-rig-smooth/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00S7: focused-seeds/geometry/S9-restricted-rig-smooth/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GAF: sources/stacks-restricted/pinned-original-source.tex, source lines 560–560. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GAJ: sources/stacks-restricted/pinned-original-source.tex, source lines 843–843. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A6A: focused-seeds/geometry/S9-restricted-rig-smooth/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0G9H: focused-seeds/geometry/S9-restricted-rig-smooth/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0G9K: focused-seeds/geometry/S9-restricted-rig-smooth/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 087R: focused-seeds/geometry/S9-restricted-rig-smooth/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let A be a Noetherian ring and let I be an ideal. Let B be a finite type A-algebra.*

1. *If $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is smooth over $\mathrm{Spec}(A) \setminus V(I)$, then $B^\wedge$ is rig-smooth over (A, I).*
2. *If $B^\wedge$ is rig-smooth over (A, I), then there exists $g \in 1 + IB$ such that $\mathrm{Spec}(B_g)$ is smooth over $\mathrm{Spec}(A) \setminus V(I)$.*

Proof. We will use Lemma 0GAJ without further mention.

Assume (1). Recall that formation of $NL_{B/A}$ commutes with localization, see Algebra, Lemma 00S7. Hence by the very definition of smooth ring maps (in terms of the naive cotangent complex being quasi-isomorphic to a finite projective module placed in degree 0), we see that $NL_{B/A}$ satisfies the fourth equivalent condition of More on Algebra, Lemma 0G9K with respect to the ideal IB (small detail omitted). Since $NL^\wedge_{B^\wedge/A} = NL_{B/A} \otimes_B B^\wedge$ by Lemma 0GAF we conclude (2) holds by More on Algebra, Lemma 0G9H.

Assume (2). Choose a presentation $B = A[x_1, \ldots, x_n]/J$, set $N = J/J^2$, and consider the element $\xi \in \mathrm{Ext}^1_B(NL_{B/A}, J/J^2)$ determined by the identity map on J/J^2. Using again that $NL^\wedge_{B^\wedge/A} = NL_{B/A} \otimes_B B^\wedge$ we find that our assumption implies the image

$$\xi \otimes 1 \in \mathrm{Ext}^1_{B^\wedge}(NL_{B/A} \otimes_B B^\wedge, N \otimes_B B^\wedge) = \mathrm{Ext}^1_{B^\wedge}(NL_{B/A}, N) \otimes_B B^\wedge$$

is annihilated by I^c for some integer $c \geq 0$. The equality holds for example by More on Algebra, Lemma 0A6A (but can also easily be deduced from the much simpler More on Algebra, Lemma 087R). Thus $M = I^c B\xi \subset \mathrm{Ext}^1_B(NL_{B/A}, N)$ is a finite submodule which maps to zero in $\mathrm{Ext}^1_B(NL_{B/A}, N) \otimes_B B^\wedge$. Since $B \to B^\wedge$ is flat this means that $M \otimes_B B^\wedge$ is zero. By Nakayama's lemma (Algebra, Lemma 00DV) this means that $M = I^c B\xi$ is annihilated by an element of the form $g = 1 + x$ with $x \in IB$. This implies that for every $b \in I^c B$ there is a B-linear dotted arrow making the diagram commute

$$\begin{array}{ccc} J/J^2 & \longrightarrow & \bigoplus B\mathrm{d}x_i \\ \downarrow{\scriptstyle b} & & \vdots{\scriptstyle h} \\ J/J^2 & \longrightarrow & (J/J^2)_g \end{array}$$

Thus $(NL_{B/A})_{gb}$ is quasi-isomorphic to a finite projective module; small detail omitted. Since $(NL_{B/A})_{gb} = NL_{B_{gb}/A}$ in $D(B_{gb})$ this shows that B_{gb} is smooth over $\mathrm{Spec}(A)$. As this holds for all $b \in I^c B$ we conclude that $\mathrm{Spec}(B_g) \to \mathrm{Spec}(A)$ is smooth over $\mathrm{Spec}(A) \setminus V(I)$ as desired. □